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Editor-in-Chief
IEEE Transactions on Smart Grid

Dear Editor,

We submit for your consideration the manuscript:

**Hierarchical Wide-Area Protection Coordination for Multi-Substation IBR
Networks: Delayed Consensus and Scalability Analysis**

for publication in *IEEE Transactions on Smart Grid*.

Context. The Self-Adaptive Model-Based Protection System (SAMBPS) achieves $P_D \geq 0.99$ for all protection functions within a single-substation domain. The thesis explicitly identifies wide-area multi-substation extension (G4) as a high-priority open problem: the single-substation SAMBPS cannot resolve fault location across substation boundaries, and standard GOOSE-based coordination cannot cope with the 10–50 ms propagation delays of wide-area networks. This paper addresses that gap.

Key contributions.

1. **Hierarchical three-layer architecture.** Local SAMBPS coordinators (Layer A, 20 ms cycle) report to regional coordinators (Layer B, 50 ms consensus window), which aggregate to a system-operator interface (Layer C). Spectral clustering of substations into K regions gives $\mathcal{O}(N \log N)$ WA-GOOSE overhead.
2. **Delayed κ_n -weighted consensus protocol.** Each substation's WA-GOOSE packet is weighted by $w_s = \kappa_n^{-1}(s) e^{-\tau_s/T_c}$ — rewarding well-conditioned LM estimates and penalising stale arrivals. This is the first WAPC protocol that exploits model-quality metrics for inter-substation message weighting.
3. **Convergence proposition.** Proves correct zone identification whenever (a) all delays $\tau_s \leq T_c = 50$ ms and (b) at least one substation in the faulted cluster achieves $\kappa_n < 30$. The proof uses graph-theoretic weight dominance, preserving the SAMBPS single-substation guarantee.
4. **Scalability validation.** 15 000 Monte Carlo trials (IEEE 39, IEEE 118, synthetic 50-substation) confirm zone accuracy 94.1–98.2%, false isolation 0.8–2.8%, and consensus convergence 45–94 ms (below the 100 ms budget). Communication overhead remains below 40 Mbps WAN capacity for $N \leq 200$.

Relationship to companion papers. This is gap paper G4 in the SAMBPS series. The SAMBPS architecture is introduced in a companion paper submitted to this journal (J4: unified five-layer architecture, C16–C20) and validated on commercial hardware in a companion submitted to *IEEE Transactions on Power Delivery* (G1: SEL-411L HIL). The present paper extends the framework to multi-substation wide-area networks, a dimension not addressed in either companion.

Suggested reviewers.

- Prof. Arun Phadke, Virginia Tech (synchrophasors, WAPC)
- Prof. Mladen Kezunovic, Texas A&M (wide-area protection, PMU)
- Prof. Ali Hooshyar, University of Toronto (IBR protection)
- Dr. Evangelos Farantatos, EPRI (wide-area monitoring, protection)

The manuscript has not been submitted elsewhere.

Sincerely,

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